



# Programmable Locking Cells (PLC) for Modular Robots with High Stiffness Tunability and Morphological Adaptability

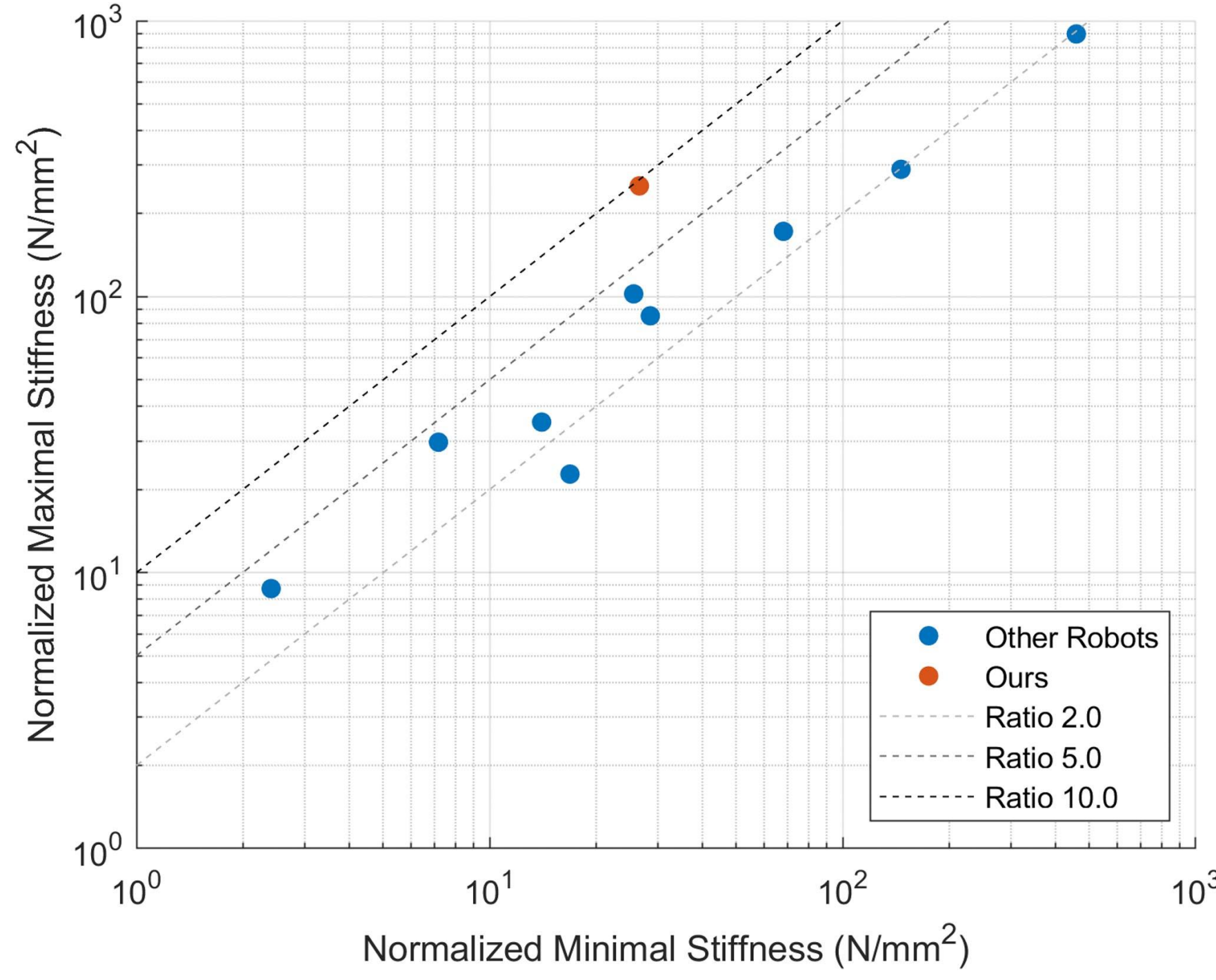
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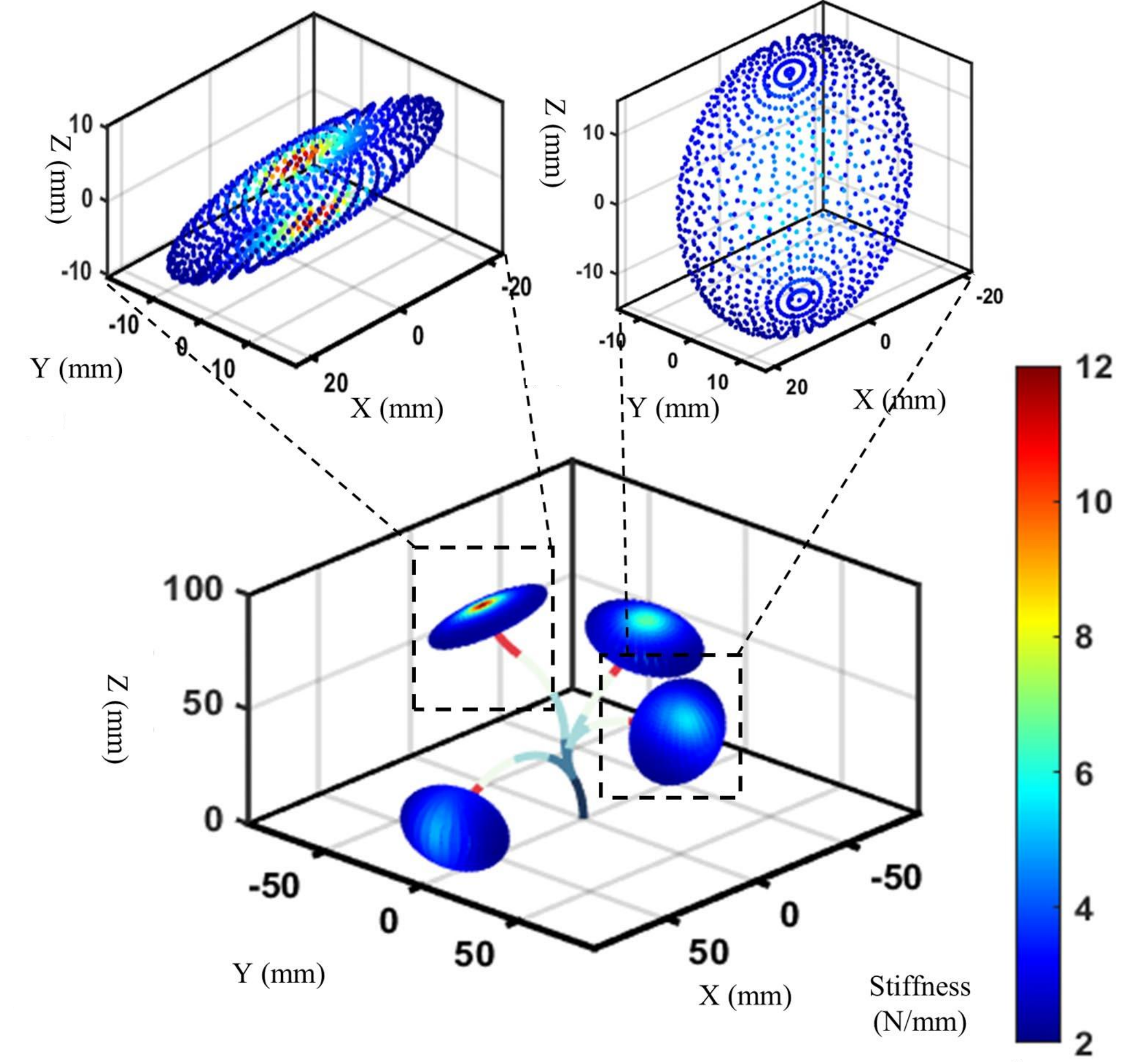
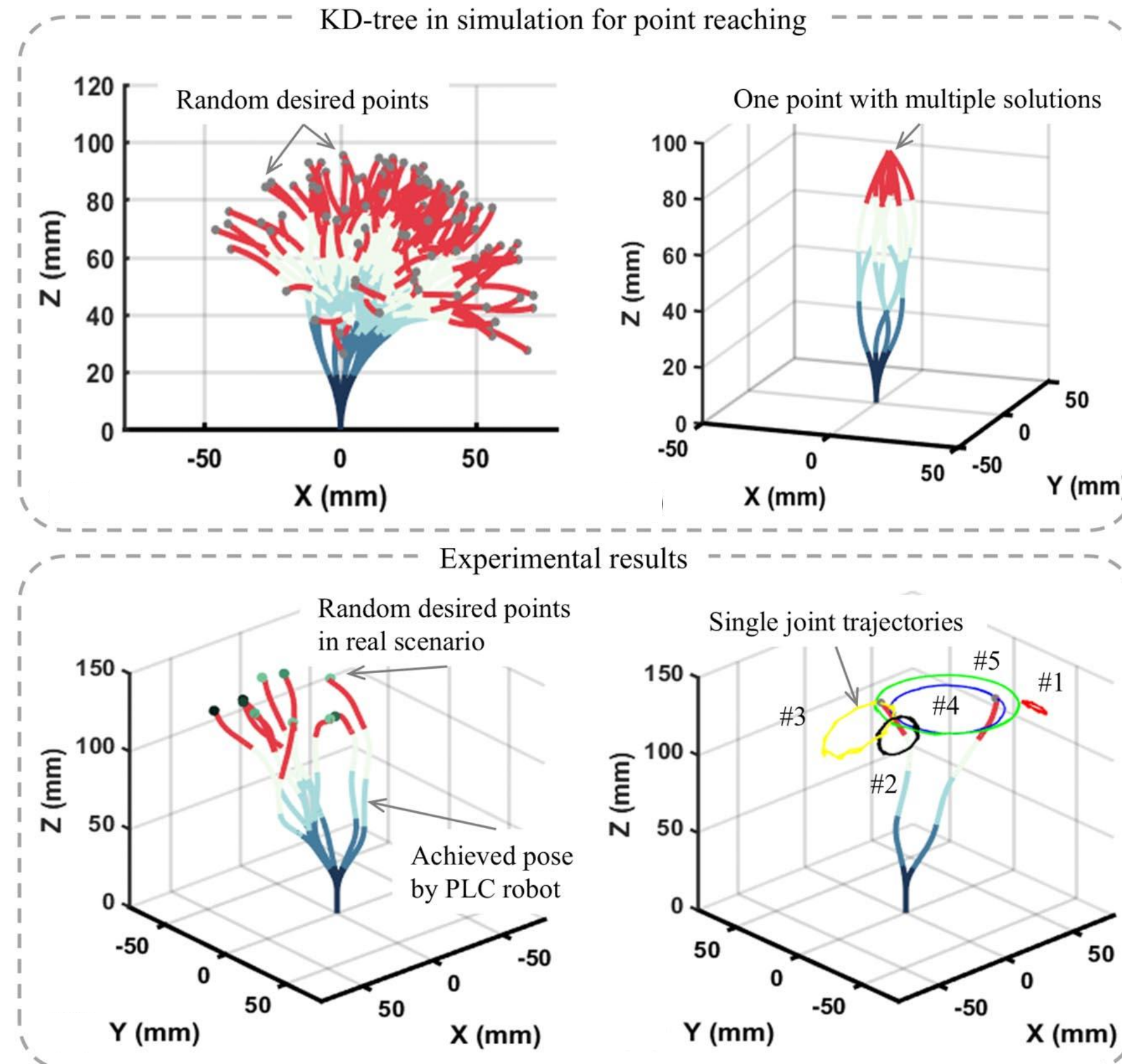


## Motivation

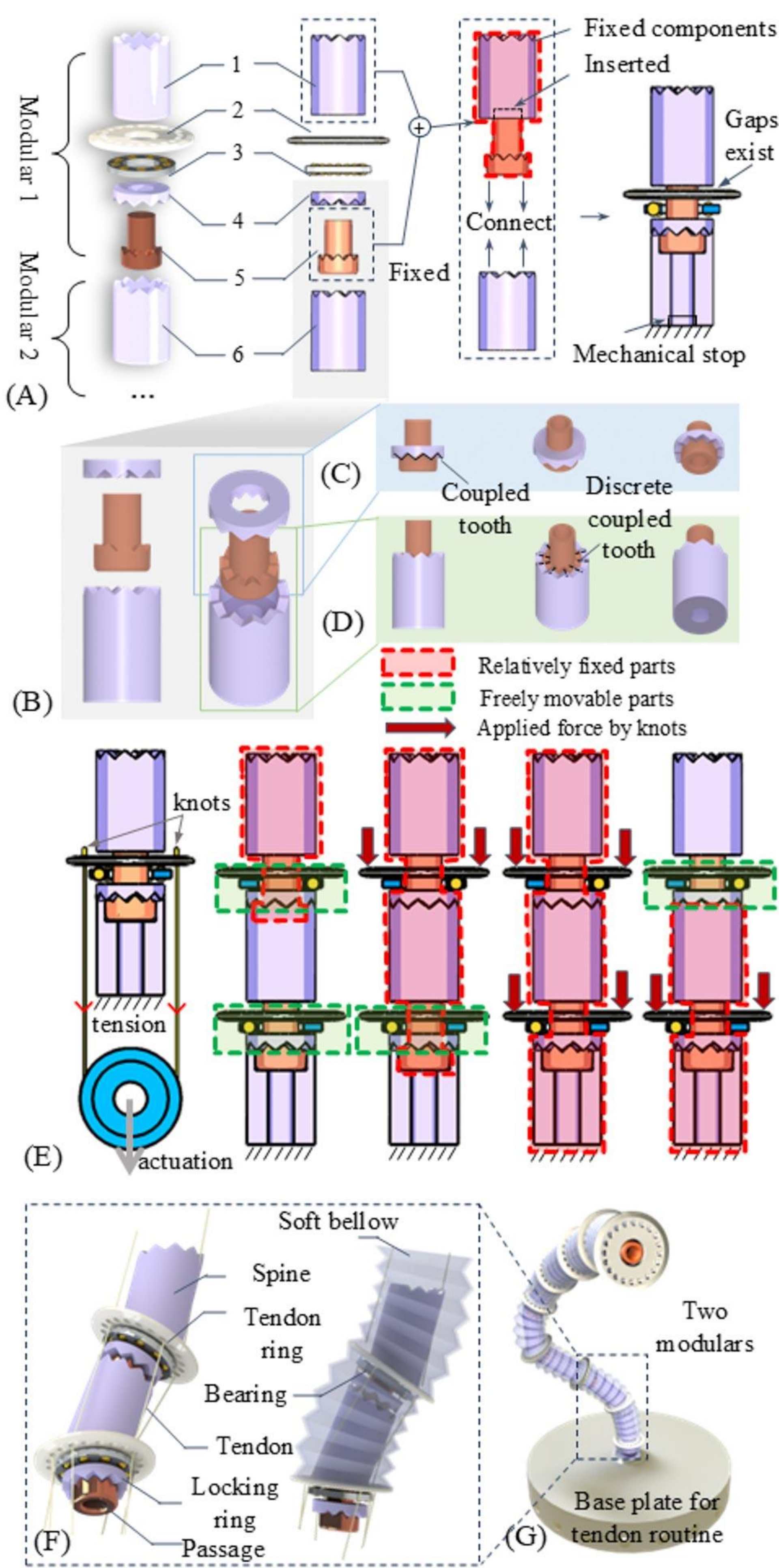
**Key idea — the Programmable Locking Cell (PLC):** a modular, tendon-driven unit that locks through mechanically interlocked teeth. **950% stiffness variation per unit.**



## Kinematics Analysis and Task Space Planning



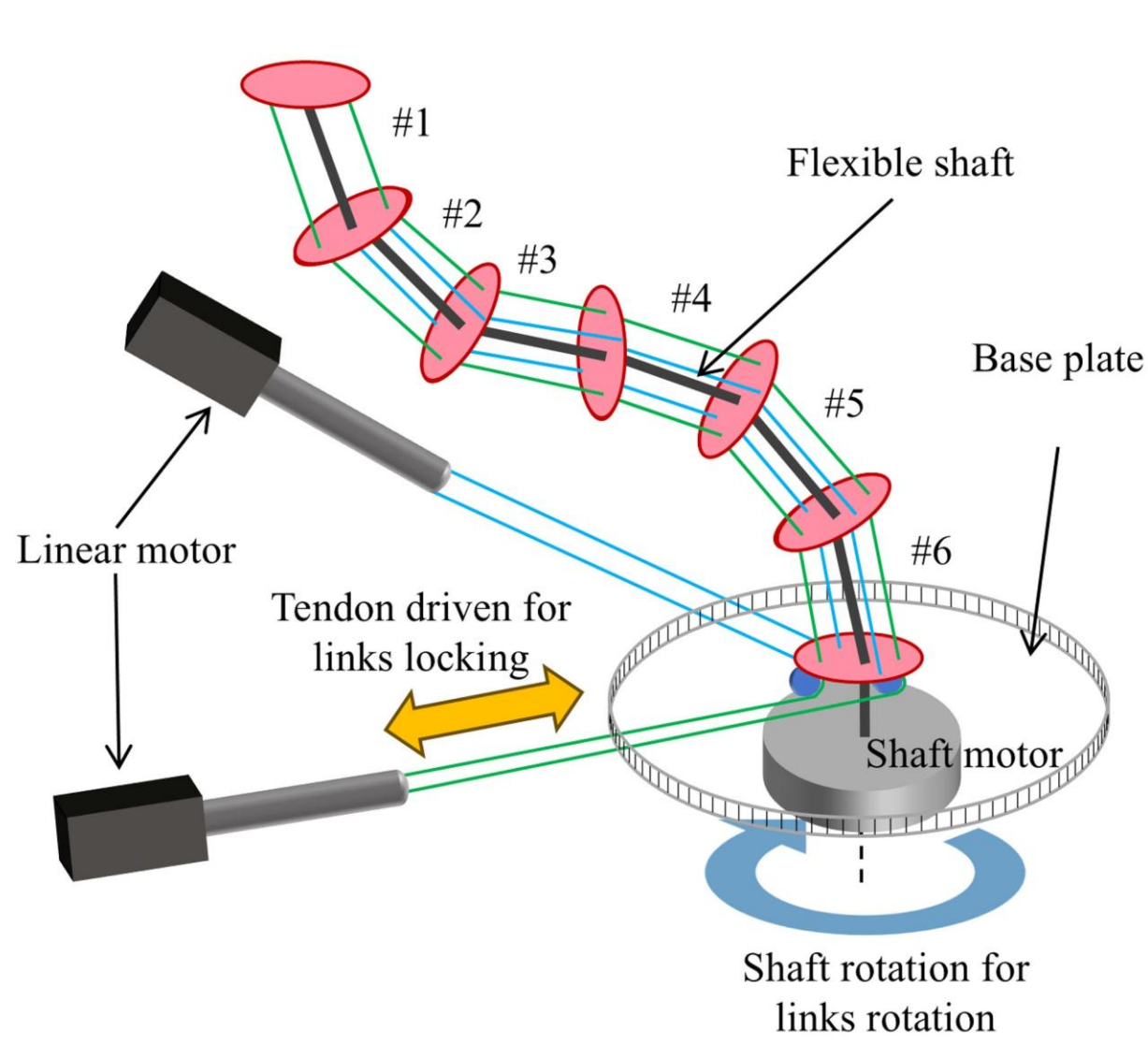
## PLC Locking Mechanism



PLC unit composition and tooth-coupling mechanism (A–D), tendon-driven locking states (E), and assembled modules (F–G).

In Subfigure (A)

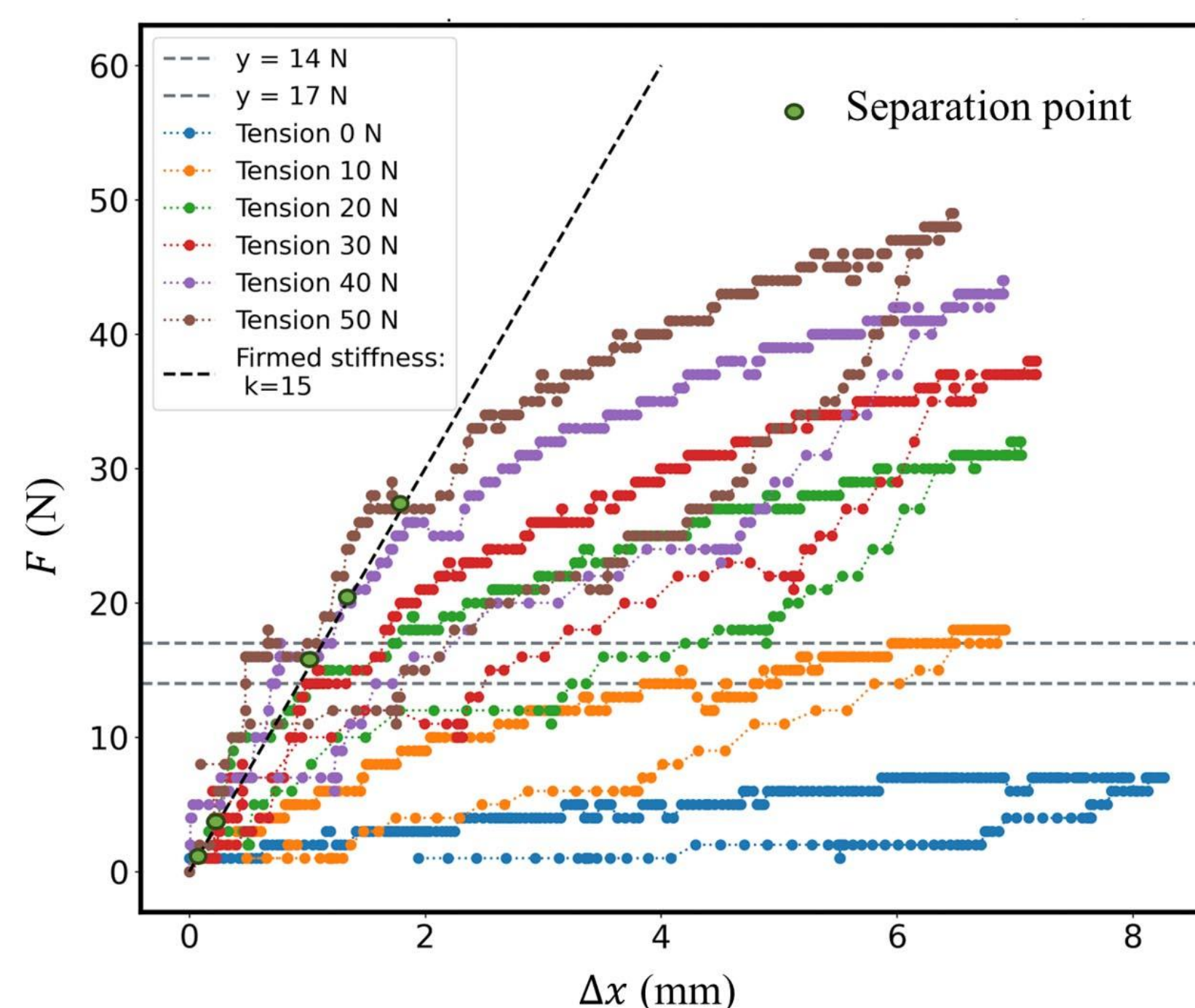
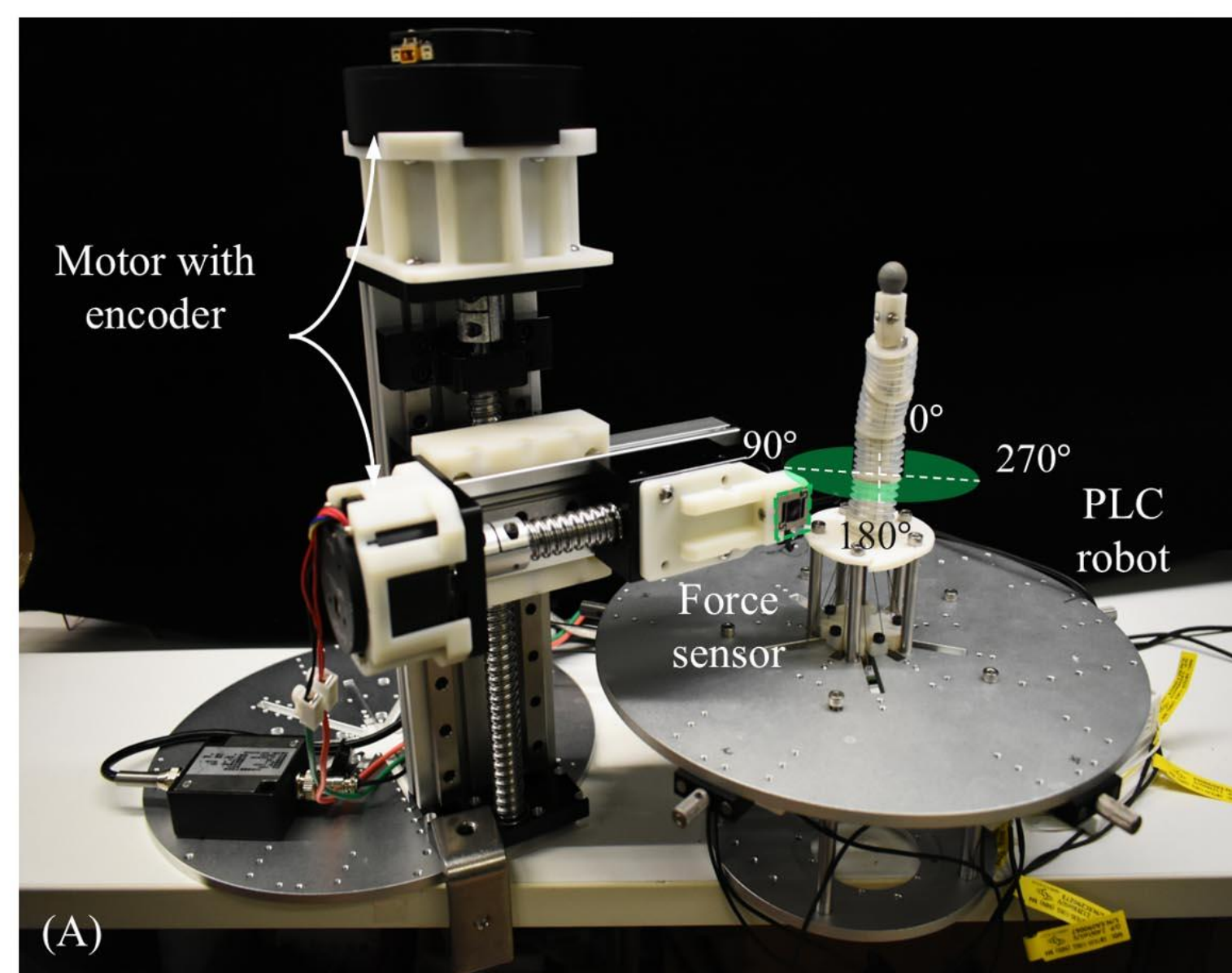
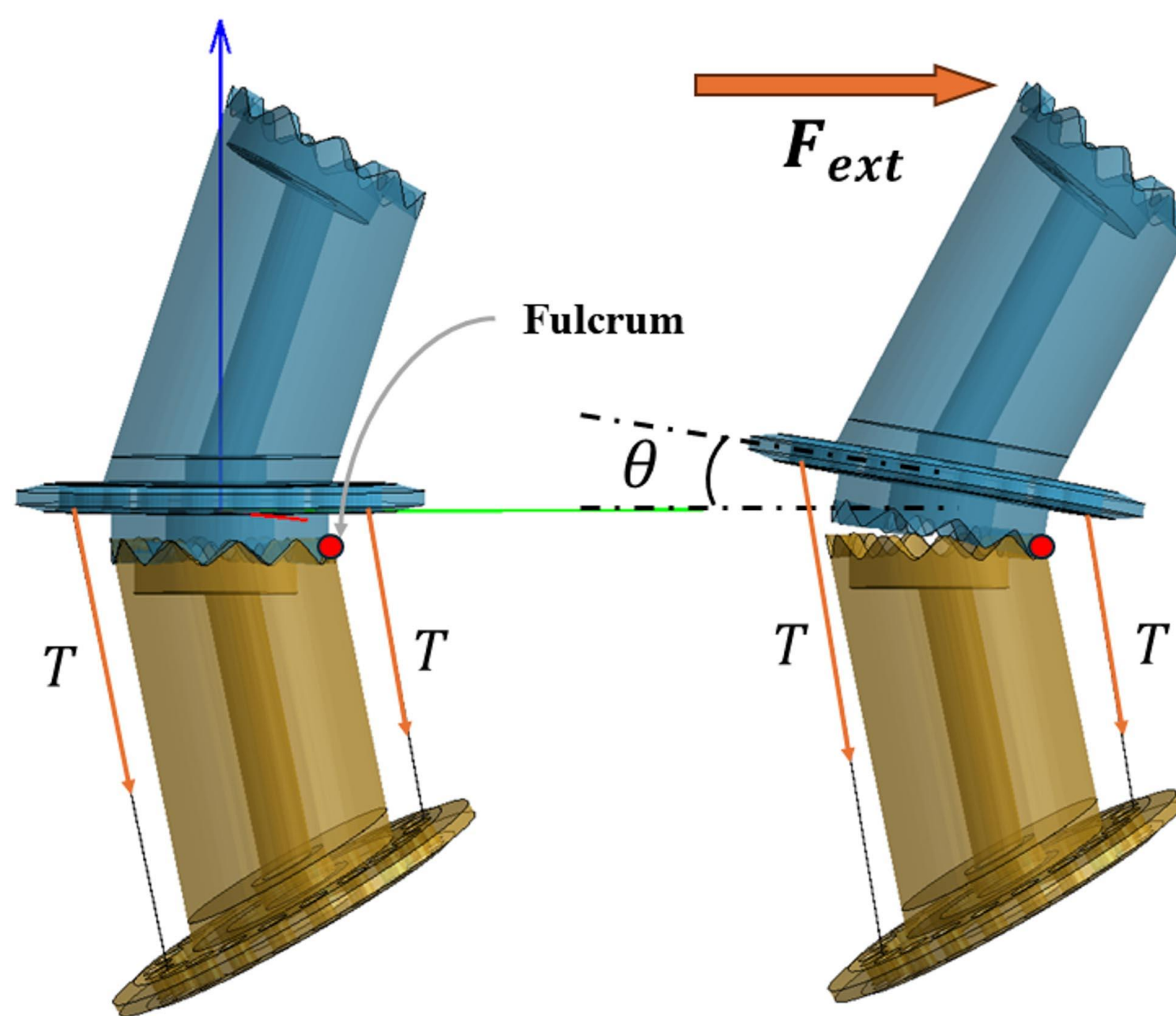
1. Upper Spine
2. Tendon Ring
3. Bearing
4. Tendon
5. Locking Ring
6. Upper Spine



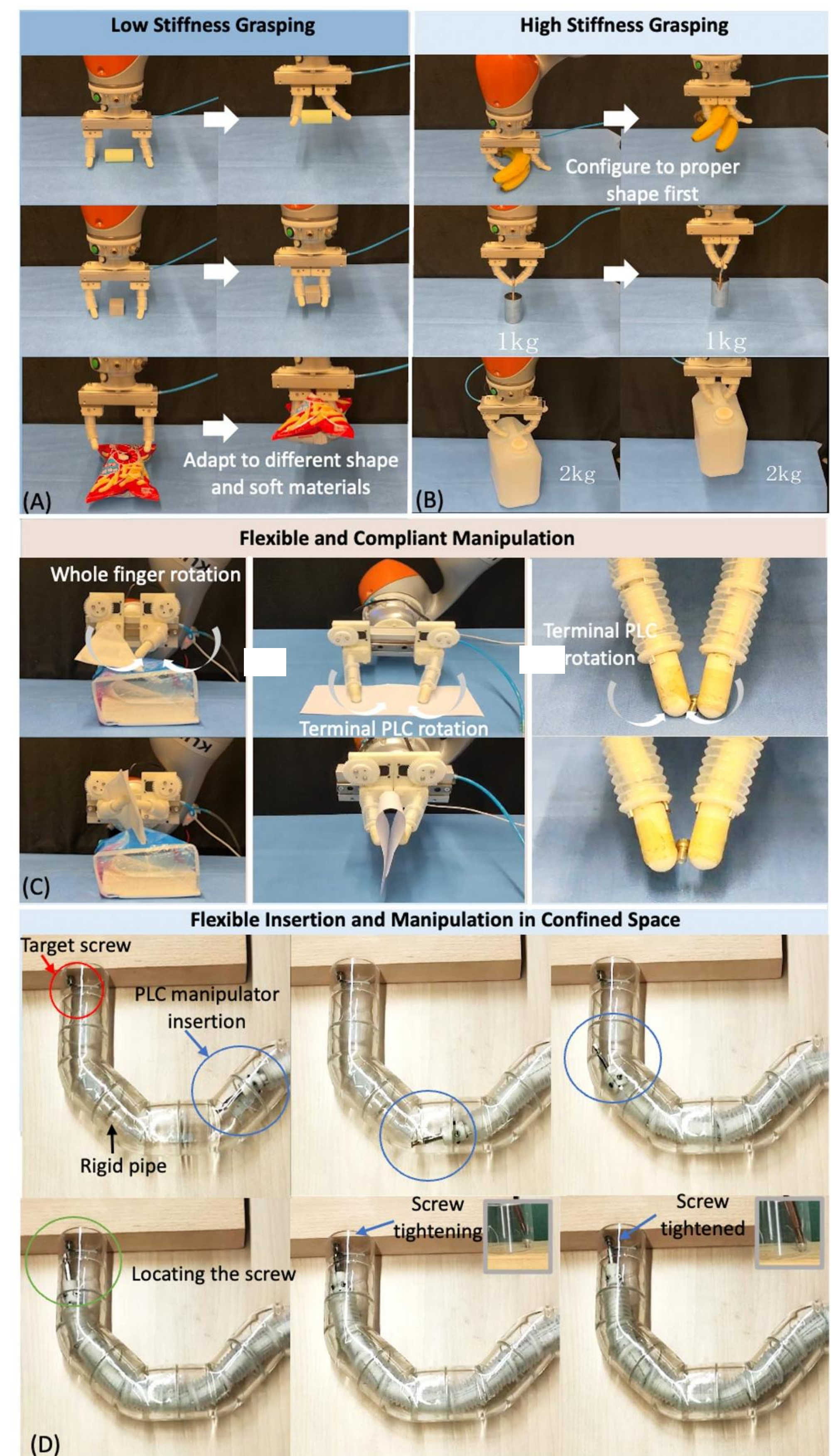
## Stiffness Modeling and Calibration

Ring detachment acts as a lever about the fulcrum  $p_0$  on the locking-ring edge; tension sets the detachment threshold but not the post-detachment slope:

$$\|\tau_{ext}\| > \|\tau_{res}\| = 2\|p_0 \times T\| \Rightarrow F_{th} \approx 0.6\|T\|$$



## Functional Demonstrations



Adaptive grasping (A), high-stiffness grasping (B), in-hand manipulation (C), pipe insertion with distal screwing (D).